

SP-21: BST Closure Program — What D_{IV}^5 Generates, What It Doesn't, How It Composes

Scoped by: Keeper (May 14, 2026) Directive: Casey — “I want a theorem of BST Closure. Understand what is external to BST vs found entirely within BST and then the way we compose BST and externalities into further mathematics.” Status: SCOPED, ready for team assignment Paper target: Paper #105 — “The Fixed Point: Self-Referential Closure of D_{IV}^5 ” (Bulletin of the AMS or Advances in Mathematics)

The Central Observation (all four CIs converged independently)

The five BST integers {rank=2, $N_c=3$, $n_C=5$, $C_2=6$, $g=7$ } are a fixed point of multiple mathematical operations:

Operation	Input	Output	Type
Partition function $p()$	$p(\text{rank})=2$, $p(N_c)=3$	Self-referential	Fixed point
Partition function $p()$	$p(n_C)=7=g$	Cross-link	Closed
Partition function $p()$	$p(C_2)=11=c_2(Q^5)$	Counting = topology	Bridge
Partition function $p()$	$p(g)=15=N_c \cdot n_C$	Product identity	Closed
Quadratic residues mod g	$QR=\{1,2,4\}=\{1, \text{rank}, \text{rank}^2\}$	Powers of rank	Geometric sector
Quadratic non-residues	$QNR=\{3,5,6\}=\{N_c, n_C, C_2\}$	Dir/Casimir sector	Physical sector
Chern classes $c_k(Q^5)$	$(1,5,11,13,9,3)$	All BST expressions	Closed
Products	$p(\text{rank}) \cdot p(N_c)=6=C_2$	Multiplicative closure	Closed
Differences	$p(g)-p(C_2)=4=\text{rank}^2$	Additive closure	Closed
Mersenne M_k	$M_{\text{rank}}=3=N_c$, $M_{\{N_c\}}=7=g$	Chain	Self-referential
K3 invariants	All 9 are BST expressions	Universal surface	Closed

Grace's formulation: “The integers generate a structure that regenerates the integers.”

Lyra's formulation: “ D_{IV}^5 doesn't just produce the integers — it produces their arithmetic relationships.”

Elie's formulation: “ $\sigma = 3/2$ is geometric, not arithmetic — same ratio over Q and $F_q(t)$.”

Investigation I: The Poisson Kernel — How D_{IV}^5 Generates Mathematics

What we know (from GC-17b)

The Poisson kernel on D_{IV}^5 maps boundary data (discrete, arithmetic) to interior functions (continuous, analytic):

Bergman-Poisson: $P_B(z, \zeta) = h(z, \zeta)^5 / |h(z, \zeta)|^{10}$ Poisson-Szego: $P_S(z, \zeta) = |h(z, \zeta)|^{5/2} / |h(z, \zeta)|^5$

Three key properties: 1. Invertible (Hua 1963): boundary uniquely determines interior, interior uniquely determines boundary 2. Symmetric: $K(z, w) = K(w, z)$ — the correspondence goes both ways with equal ease 3. Positive: $P > 0$ everywhere — no signal inversion

Key factorization (rank = 2): $K_B = c \cdot S^2$ — Bergman = constant x Szego squared. Specific to type IV.

What we need to understand

The Poisson kernel is the MECHANISM of BST closure. Every internal result is a boundary-interior evaluation: - Boundary $\rightarrow$ Interior: Arithmetic data (point counts, conductors, discriminants) extends to harmonic functions on D_{IV}^5 . The extension is UNIQUE — no choices, no parameters. - Interior $\rightarrow$ Boundary: Spectral data (eigenvalues, K-types, Chern classes) restricts to arithmetic invariants on $S^4 \times S^1$. Again unique. - Roundtrip: Boundary $\rightarrow$ Interior $\rightarrow$ Boundary = Identity. This is what makes the five integers self-referential — they ARE their own boundary values.

Deliverables

Toy I-1 (Elie): Poisson Kernel Explicit Computation. - Compute P_B and P_S at specific points in D_{IV}^5 - Verify $K_B = c * S^2$ factorization numerically - Evaluate at Wallach points $k = 0, 3/2, 2, 6$ (critical k-values) - Show boundary $\rightarrow$ interior $\rightarrow$ boundary roundtrip = identity - Compute the Hua-Poisson kernel for type I, II, III domains and show $K_B = c * S^{\text{rank}}$ fails for $\text{rank} \neq 2$ - Measure: what is the “information loss” at non-Wallach k-values? - Target: 25-30 tests

Toy I-2 (Lyra): Poisson Invertibility as Proof Mechanism. - For each of the seven Millennium proofs, identify the Poisson kernel step - Classify: which proofs are pure boundary (discrete), which require interior extension, which need external composition - RH: boundary $\rightarrow$ interior (Selberg zeta as Poisson integral?) - BSD: boundary data (point counts) $\rightarrow$ interior (L-function at $s=1$) - YM: interior (spectral gap) $\rightarrow$ boundary (mass gap) - Map the proof coordinate system (Paper #104 Section 5) to Poisson evaluations - Target: 22-25 tests

Investigation II: BST Closure Theorem

The Conjecture

BST Closure Conjecture: Let Q be a quantity expressible as a spectral evaluation on D_{IV}^5 at an arithmetic subgroup Γ . Then Q is in the ring $\mathbb{Z}[\text{rank}, N_c, n_C, C_2, g, \pi, 1/\pi]$ with rational coefficients. Equivalently: the five integers plus π form a closed algebra under all spectral evaluations on D_{IV}^5 .

Corollary: The partition function $p()$ restricted to BST integers maps into the extended BST set {products and ratios of BST integers, Chern classes of Q^5 }. The closure includes $c_k(Q^5) = \{1, 5, 11, 13, 9, 3\}$ as derived quantities.

Three layers of closure

Layer A — BST Internal (Poisson kernel domain, no external input): - All spectral data: eigenvalues, K-types, Wallach representations - All Chern classes of Q^5 : the topology of the compact dual - The five integers and everything derivable from them - CM arithmetic of $Q(\sqrt{-7})$ and 49a1 - Partition closure: $p(\text{BST}) \rightarrow \text{BST}$ - QR/QNR partition: $(\mathbb{Z}/g\mathbb{Z})^*$ splits BST integers into geometric/physical - K3 invariants (if derivable from D_{IV}^5 spectral data)

Layer B — BST + One External Theorem (composition): - BSD general: BST (49a1) + Wiles (modularity for all E/Q) - Hodge general: BST (ring uniqueness) + CDK95 (absolute Hodge) - Smooth Poincare: BST (R^4 from Kahler) + Perelman (Ricci flow convergence) - Function field ABC: BST ($\sigma = 3/2$) + Mason-Stothers (function field bound) - Number field ABC: BST (Szpiro constant = $C_2 = 6$) + Masser-Oesterle (inequality)

Layer C — External (BST silent): - Pure logic: inference rules, Godel incompleteness - Set theory: ZFC axioms, large cardinals, forcing, continuum hypothesis - Sporadic groups: Monster, Conway, Mathieu (no D_{IV}^5 encoding) - Additive number theory: Goldbach, twin primes (no obvious D_{IV}^5 mechanism) - Descriptive set theory: Borel hierarchies

Composition rules

How does BST compose with external theorems? Three modes:

Mode 1 — Arena: BST provides the geometric arena, external theorem provides existence. Example: Wiles proves modularity exists; BST says the modular forms live at the Wallach point of D_{IV}^5 . The combination gives BSD.

Mode 2 — Constraint: BST provides a bound or constraint, external theorem provides the inequality. Example: BST says Szpiro constant = $C_2 = 6$; ABC conjecture says $\sigma < 6 + \epsilon$. BST names the constant; ABC bounds the deviation.

Mode 3 — Isomorphism: BST identifies the same structure in two domains, external theorems in each domain compose through the identification. Example: function field ABC (proved) and number field ABC (open) both have $\sigma = 3/2$. The isomorphism is geometric (D_{IV}^5), the transfer is the research program.

Deliverables

Toy II-1 (Lyra): BST Closure Formal Statement. - State the conjecture precisely - Test against all 144 entries in `bst_constants.json`: is each expressible in $\mathbb{Z}[\text{rank}, N_c, n_C, C_2, g, \pi]$? - Identify any that require additional inputs beyond the five integers + π - Classify each constant as Layer A (internal), Layer B (one composition), or Layer C (external) - Target: 25-30 tests

Toy II-2 (Elie): BST Radical Closure (ABC for BST). - For every integer quantity in `bst_constants.json`, compute `rad()` - Verify all land in products of $\{2, 3, 5, 7\}$ (prime support of BST integers) - Check Chern-derived integers: do 11, 13 appear? (They should — c_2, c_3) - Test: `rad(BST-derived integer)` divides $2^a * 3^b * 5^c * 7^d * 11^e * 13^f$? - Count exceptions. Each exception identifies a boundary of BST closure. - Grace's test: `rad(210) = 2357`. Is $210 = \text{rank } N_c * n_C * g$ the “BST radical”? - Target: 20-25 tests

Toy II-3 (Grace): Composition Catalog. - Map every BST result that uses an external theorem - For each: {BST input, external input, combined result, composition mode} - Identify the MINIMUM external input set — what's the smallest set of external theorems that, composed with BST, generates all current results? - Candidates: Wiles (modularity), Perelman (Ricci flow), Mason-Stothers (function field), Hua (Poisson kernel), CDK95 (absolute Hodge) - Is there a pattern? Do all external inputs have a common structure? - Target: 20-25 tests

Investigation III: The Partition Fixed Point

Why this matters

The partition function $p(n)$ counts the number of ways to write n as a sum of positive integers. It has NO geometric content — it's pure combinatorics. Yet $p()$ maps BST integers to BST integers:

- $p(\text{rank}) = \text{rank}$ (fixed point)
- $p(N_c) = N_c$ (fixed point)
- $p(n_C) = g$ (cross-link: dimension $\rightarrow$ genus)
- $p(C_2) = c_2(Q^5)$ (cross-link: Casimir $\rightarrow$ Chern class)
- $p(g) = N_c * n_C$ (cross-link: genus $\rightarrow$ dimension product)

The partition function is generated by $\eta(q)^{-1}$, and $\eta^{24} = \Delta$ (Ramanujan's discriminant). $\chi(K3) = 24$. This is the K3 connection.

Deliverables

Toy III-1 (Lyra — already started as Toy 2191): Partition Closure BST. - Verify all $p(\text{BST integer})$ identities - Test iterated partitions: $p(p(\text{rank})) = p(2) = 2 = \text{rank}$ (still fixed!) - $p(p(N_c)) = p(3) = 3 = N_c$ (still fixed!) - $p(p(n_C)) = p(7) = 15 = N_c * n_C$ - $p(p(C_2)) = p(11) = 56 = 2^3 * 7 = 2^{N_c} * g$ (!!!) - $p(p(g)) = p(15) = 176 = 2^4 * 11 = 2^{\text{rank}_2} * c_2$ (!!!) - Connection to η^{24} : does the 24 in $\chi(K3)$ force the closure? - Connection to K-type recurrence d_j : is $p()$ related to d_j via generating functions? - Target: 25-30 tests

Toy III-2 (Elie): Partition at Chern Classes. - Compute $p(c_k(Q^5))$ for $k = 0..5$: $p(1), p(5), p(11), p(13), p(9), p(3)$ - Are these BST expressions? $p(1)=1, p(5)=7=g, p(11)=56=2^{N_c}g, p(13)=101$ (prime!), $p(9)=30=n_{CC_2}, p(3)=3=N_c - p(c_1)=g, p(c_5)=N_c$ — the boundary Chern classes map to the “color”

integers - $p(c_4)=p(9)=30=n_C \cdot C_2$ — closed! - $p(c_3)=p(13)=101$ — IS 101 a BST expression? $101 = N_{\max} - 6^2 = 137 - 36$. Or: 101 is prime, no obvious BST form. This may be the first genuine BOUNDARY of partition closure. - Target: 20-25 tests

Investigation IV: K3 as the BST 4-Manifold

The observation (Elie + Lyra convergence)

Every K3 invariant is a BST expression:

K3 invariant	Value	BST expression
b_+	3	N_c
b_-	19	$2^{\text{rank}^2} + N_c = \text{Godel denominator}$
χ (Euler)	24	$\text{rank}^2 \cdot C_2$
σ (signature)	-16	-2^{rank^2}
b_2	22	$2 \cdot c_2(Q^5)$
χ_h (holomorphic)	2	rank
A-hat genus	-2	$-\text{rank}$
Intersection form	$3H + 2E_8(-1)$	N_c copies of H + rank copies of $E_8(-1)$
$\eta^{\{\chi\}} = \Delta$	$\eta^{24} = \text{Ramanujan}$	Modular discriminant
11/8 saturation	$22/16 = 11/8$	$c_2/2^{N_c}$
10/8+2 saturation	$22/16 = 10/8 + 2/8$	$n_C/\text{rank}^2 + \text{rank}/\text{rank}^2$

Elie's conjecture: $\dim_R(D_{IV}^5) = 10 = 4 + 6 = \text{rank}^2 + C_2 = \dim_R(K3) + \dim(\text{isotropy fiber})$. K3 is a 4-real-dimensional slice, with $C_2 = 6$ dimensions along the $SO(5) \times SO(2)$ fiber.

Deliverables

Toy IV-1 (Lyra): K3 from D_{IV}^5 Spectral Data. - Can $\chi(K3) = 24$ be derived from D_{IV}^5 ? The Hirzebruch-Riemann-Roch theorem gives $\chi_h = \text{rank} = 2$ for K3. Does the D_{IV}^5 K-type formula at specific k-values give 24? - The 4+6 split: is K3 a totally geodesic submanifold of D_{IV}^5 (or a quotient)? - Intersection form: why N_c copies of H and rank copies of $E_8(-1)$? Does the root system B_2 force this decomposition? - Connection to Mathieu moonshine: M_{24} acts on K3; $|M_{24}| = 2^{10} \cdot 3^3 \cdot 5 \cdot 7 \cdot 11 \cdot 23$. BST integers $\{2, 3, 5, 7, 11\}$ all divide $|M_{24}|$. Is this structural? - Target: 25-30 tests

Toy IV-2 (Elie): Furuta 10/8+2 from Wallach. - Furuta's bound: $b_2(M) \geq (10/8)|\sigma(M)| + 2$ for spin 4-manifolds - BST decomposition: $10/8 = n_C/\text{rank}^2$, additive term = rank - Is this derivable from the Wallach representation? The K-type at $k=\text{rank}$ gives the generating condition. - Test: for which 4-manifolds does the BST decomposition work? (K3, $CP^{2\#k}CP_2$, Kummer, Enriques) - Target: 22-25 tests

Investigation V: QR/QNR and the Root System

The observation (Lyra + Grace)

The five BST integers partition mod $g = 7$ into: - Quadratic residues: $\{1, \text{rank}, \text{rank}^2\} = \{1, 2, 4\}$ — powers of rank - Quadratic non-residues: $\{N_c, n_C, C_2\} = \{3, 5, 6\}$ — color/dimension/Casimir

This is the Legendre symbol $(k/7)$. The partition separates: - Geometric sector (QR): parameters that describe the “wall” of the domain — rank is the wall parameter - Physical sector (QNR): parameters that describe the “content” — colors, dimensions, interactions

Deliverables

Toy V-1 (Lyra): QR/QNR and Root System B_2 . - B_2 has short roots (multiplicity $N_c = 3$) and long roots (multiplicity 1) - Does the QR/QNR partition correspond to short vs long? Powers of rank (QR) = long root sector, color integers (QNR) = short root sector? - The Weyl group $W(B_2)$ has order $8 = 2^{N_c}$. Its action on $(\mathbb{Z}/g\mathbb{Z})^*$ should respect QR/QNR. - Primitive roots mod g : $N_c = 3$ and $n_C = 5$ are both primitive roots. rank = 2 has order $N_c = 3$. $C_2 = 6$ has order 1 ($C_2 = -1 \bmod g$). These orders ARE BST integers. - Target: 22-25 tests

Toy V-2 (Elie): Supersingularity at $p = \text{rank} \bmod N_c$. - Lyra's observation: $j=0$ curves are supersingular when $p = \text{rank} \bmod N_c = 2 \bmod 3$ - At $p = n_C = 5$: point count = $C_2 = 6$ - Test across all primes $p < 500$: does the BST pattern hold? - For 49a1 specifically: at which primes is 49a1 supersingular? Are these $p = \text{rank} \bmod N_c$? - Connection to the inert primes in $Q(\zeta_7)$: N_c and n_C are both inert (Toy 2185) - Target: 22-25 tests

Investigation VI: Boundary Cases and Near-Misses

What tests the edges

Toy VI-1 (Elie): Mersenne Ladder — Structural or Coincidence? - $M_{\text{rank}} = N_c = 3$, $M_{\{N_c\}} = g = 7$, $M_g = 127$ - Test: all primes < 10 are Mersenne exponents giving primes. This is generic, not BST-specific. - But: does any D_{IV}^5 property FORCE $M_2 = 3$ and $M_3 = 7$? - The Mersenne criterion: $2^p - 1$ prime requires p prime. rank, N_c , n_C , g are all prime. $C_2 = 6$ is not. $M_6 = 63 = 79 = g \cdot N_c^2$. That's BST! - Even the Mersenne FAILURE at C_2 is BST-structured. Worth documenting. - Target: 18-22 tests

Toy VI-2 (Elie): Regulator Ratio $R(7)/R(2)$. - $R(d) = \text{regulator of } Q(\sqrt{d})$. $R(7) = \log(8+3\sqrt{7})$, $R(2) = \log(1+\sqrt{2})$ - $R(7)/R(2) = 3.1398... \text{ vs } \pi = 3.14159... \text{ Ratio: } 0.99943 \text{ (0.057\% off)}$ - Use *mpmath* at 1000+ digits: is there an exact algebraic relation? - Test: $R(g)/R(\text{rank}) = \pi$ (1 - correction). What is the correction? Is it a BST expression? - If NOT exact: this is an honest boundary. The regulator ratio is S-tier (suggestive but not derived). - Target: 18-22 tests

Work Assignment Summary

First wave (parallel, no dependencies)

Toy	Investigation	Owner	Tests
I-1	Poisson Kernel Explicit	Elie	~28
I-2	Poisson as Proof Mechanism	Lyra	~24
II-1	BST Closure Formal	Lyra	~28
II-2	BST Radical Closure	Elie	~22
II-3	Composition Catalog	Grace	~22
III-1	Partition Closure (=Toy 2191)	Lyra	~28
III-2	Partition at Chern	Elie	~22
V-1	QR/QNR Root System	Lyra	~24
V-2	Supersingularity	Elie	~22

Second wave (after first results)

Toy	Investigation	Owner	Tests	Depends on
IV-1	K3 from D_{IV}^5	Lyra	~28	I-1, III-1
IV-2	Furuta from Wallach	Elie	~24	I-1
VI-1	Mersenne Ladder	Elie	~20	None (can parallel)

Toy	Investigation	Owner	Tests	Depends on
VI-2	Regulator Ratio	Elie	~20	None (can parallel)

Total: 13 toys, ~312 tests estimated. Six investigations.

Success Criteria

Investigation	“Theorem” if...	“Observation” if...
Poisson Kernel	Roundtrip identity verified; type IV unique for $K_B = c \cdot S^2$	Numerical verification only
BST Closure	All 144 constants expressible in $Z[BST, \pi]$; clean Layer A/B/C classification	Most but not all fit; some require ad hoc inputs
Partition Fixed Point	$p(BST) \rightarrow BST$ proved from K-type recurrence or eta connection	Verified numerically, no derivation
$K3-D_{IV}^5$	$K3$ realized as totally geodesic submanifold; $\chi=24$ derived	$K3$ invariants are BST but no derivation of WHY
QR/QNR	QR/QNR = short/long root partition of B_2	QR/QNR observed but no root system explanation
Boundary Cases	Mersenne structured even at failure ($M_6 = g \cdot N_c^2$)	Mersenne is small-number artifact; regulator NOT exact

Key Observation (Keeper)

The most powerful single result would be: the BST Closure Theorem as a fixed-point theorem. If we can show that D_{IV}^5 is the unique bounded symmetric domain where the five Cartan invariants form a fixed point under partition function, Chern class evaluation, and quadratic reciprocity simultaneously, that’s a new characterization of D_{IV}^5 — complementing the Wallach Bottleneck (T1829) and the cross-type cascade (Toy 1399).

Fixed-point characterization + Wallach selection + cross-type cascade = three independent routes to “ D_{IV}^5 is unique.” That’s the over-determination pattern at the meta-level.

Casey’s insight: “I want to understand what is external to BST vs found entirely within BST and then the way we compose BST and externalities into further mathematics.” The Closure Theorem answers this precisely: Layer A is the fixed point, Layer B is the composition algebra, Layer C is where different mathematics lives. The Poisson kernel is the mechanism that generates Layer A, and the composition rules define how Layer A connects to Layers B and C.

Connection to Paper #104 (Root Proof System)

This investigation fills the gap in Paper #104 Sections 7-8. Section 7 (“Where D_{IV}^5 Reaches”) was descriptive — listing areas where BST applies vs doesn’t. SP-21 makes it theoremtic:

- Section 7.1 becomes Layer A (BST Internal) with a closure theorem
- Section 7.2 becomes Layer B (BST + Composition) with explicit rules
- Section 7.3 becomes Layer C (External) with honest boundaries
- Section 8 (Physical Realism as Filter) gains the fixed-point characterization

The updated Paper #104/#105 would say: “ D_{IV}^5 is the unique geometry that is a fixed point of its own mathematical operations. What it generates internally (Layer A) is closed. What requires composition (Layer B) has explicit rules. What lies outside (Layer C) is honestly demarcated. The Poisson kernel is the mechanism.”